MultiConAD: A Unified Multilingual Conversational Dataset for Early Alzheimer's Detection

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Introduction

Background

- Dementia is a progressive cognitive syndrome with Alzheimer's disease (AD) as the leading cause. Conversation-based AD detection offers a cost-effective alternative to clinical methods, as language dysfunction is an early biomarker of AD.
- Most prior research has framed AD detection as a binary classification problem, limiting the ability to identify Mild Cognitive Impairment (MCI), a crucial stage for early intervention.
- Studies also primarily rely on single-language datasets, mainly in English, restricting cross-language generalizability.

Aim

1. Present a new multilingual conversational dataset for AD detection.
2. Perform fine-grained three-class classification (AD, MCI, HC).
3. Analysis of classification performance in monolingual and multilingual settings.

Dataset Creation

Sources

- Sixteen publicly available AD-related conversational datasets are unified to create a multilingual dataset in Chinese, Spanish, English, and Greek.
- Tables 1 and 2 provide an overview of all datasets used in this study and two sample conversation from two different sources.

Language	Source	Modality		Task					Labels			
		Text	Audio	PD	FT	SR	FC	NA	DM	AD	MCI	HC
English	Pitt	✓	✓	✓	✗	✗	✗	✗	✗	255	42	243
	Lu	✓	✓	✓	✗	✗	✗	✗	6	16	2	27
	VAS	✓	✓	✗	✗	✗	✓	✗	30	✗	35	36
	Baycrest	✓	✓	✗	✗	✓	✗	✗	✗	3	7	✗
	Kempler	✓	✓	✓	✗	✗	✗	✗	✗	7	✗	✗
	WLS	✓	✓	✓	✓	✗	✗	✗	✗	263	✗	1106
	Delware	✓	✓	✓	✗	✗	✗	✓	✗	✗	61	34
	Taukadial	✗	✓	✓	✗	✗	✗	✗	✗	✗	95	74
Spanish	Ivanova	✓	✓	✗	✗	✗	✗	✓	✗	74	90	197
	PerLA	✓	✗	✓	✓	✗	✓	✗	✗	21	✗	✗
Chineas	NCMMSE	✗	✓	✓	✓	✗	✓	✗	✗	79	93	108
	iFkyTek	✓	✗	✓	✗	✗	✗	✗	✗	68	144	111
Greek	DS3	✗	✓	✓	✓	✗	✗	✗	✗	76	✗	19
	DS5	✗	✓	✓	✓	✗	✗	✗	✗	26	35	31
	DS7	✗	✓	✓	✓	✗	✗	✗	✗	27	35	2
	ADReSS-M	✗	✓	✓	✗	✗	✗	✗	✗	22	✗	24

Table 1. Dataset information Table. Tasks: (PD) Picture Description, (FT) Fluency Task, (SR) Story Retelling, (FC) Free Conversation, (NA) Narrative. Labels: (DM) Dementia, (AD) Alzheimer's Disease, (MCI) Mild Cognitive Impairment, (HC) Healthy Control.

Dataset	Transcript
Pitt	INT: just tell me everything that you see happening in that picture. PAR: alright. there's &-um a young boy that's getting a cookie jar. PAR: and it he's &-uh in bad shape because &-uh the thing is PAR: and in the picture the mother is washin(g) dishes and doesn't see [...]
Taukadial	What I see is a young lady was riding her tricycle with her cat, and somehow the cat got up in the tree. Her dog is trying to chase the cat, can't get up the tree. Her father climbed the tree and is sitting on the limb [...]

Table 2. Two conversation samples from Pitt and Taukadial Datasets.

Dataset Creation Pipline

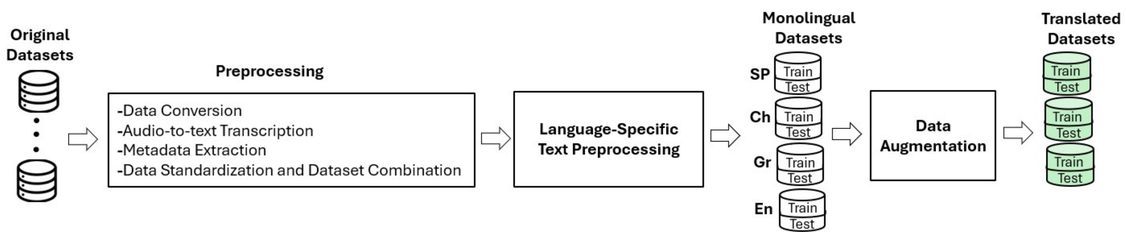

Figure 1: The flowchart illustrates the dataset creation pipeline, including data conversion, audio transcription, metadata extraction, data standardization, and language-specific text cleaning to build a unified multilingual dataset.

Language	Train Split			Test Split		
	AD	MCI	HC	AD	MCI	HC
Spanish	78	70	156	20	18	39
Chinese	93	200	187	23	51	47
Greek	266	51	87	66	13	22
English	512	239	1450	63	60	87
Total	**949**	**560**	**1880**	**172**	**142**	**195**

Table 3. Final distribution of diagnostic categories (AD, MCI, HC) across languages with corresponding train and test splits.

Experiment

Method pipeline

Two text representation methods, dense and sparse, and four machine learning classifiers are used separately to perform binary and three-class classification in the three following experimental scenarios.

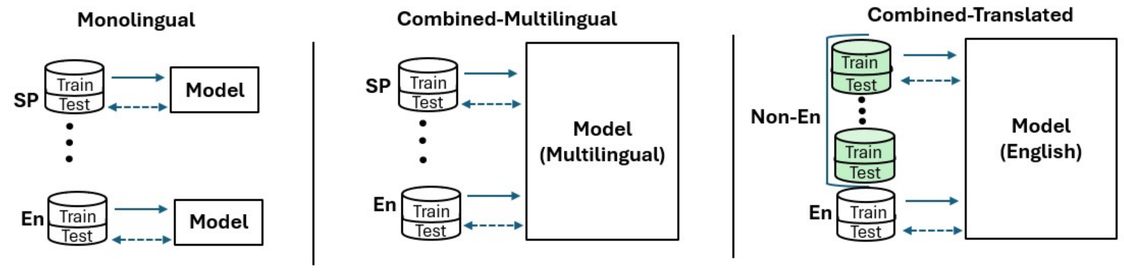

Figure 2: Flowchart depicting the experimental scenarios.

Three-class classification results

Language	Text Repr.	Monolingual				Combined-Multilingual				Combined-Translated			
		DT	RF	SVM	LR	DT	RF	SVM	LR	DT	RF	SVM	LR
Spanish	Sparse	0.61	0.60	**0.61**	0.61	0.51 ↓	**0.62** ↑	0.51 ↓	0.58 ↓	0.47 ↓	**0.58** ↓	0.56 ↓	0.56 ↓
	Dense	0.52	0.61	**0.61**	0.61	0.47 ↓	**0.61**	0.61	0.57 ↓	**0.60** ↑	0.58 ↓	0.56 ↓	0.51↓
Chinese	Sparse	0.36	0.35	**0.40**	0.39	**0.42** ↑	0.39 ↑	0.39 ↓	0.40 ↑	0.45 ↑	0.59 ↑	**0.68** ↑	0.62 ↑
	Dense	0.51	0.58	**0.59**	0.56	0.43 ↓	**0.62**↑	0.60 ↑	0.60 ↑	0.43 ↓	**0.64**↑	0.60 ↑	0.45 ↓
Greek	Sparse	0.59	**0.74**	0.67	0.71	0.57 ↓	**0.71** ↓	0.53 ↓	0.66 ↓	0.64 ↑	0.65 ↓	**0.69**↑	0.60 ↓
	Dense	0.54	0.66	**0.73**	0.73	0.54	0.66	0.65 ↓	**0.67** ↓	**0.62** ↑	0.61 ↓	0.60 ↓	0.42 ↓
English	Sparse	0.59	0.62	**0.65**	0.65	0.59	0.58 ↓	0.41 ↓	**0.66** ↑	0.50 ↓	0.61 ↓	**0.66**↑	0.64 ↓
	Dense	0.51	0.62	**0.65**	0.63	0.50 ↓	0.62	**0.65**	0.63	0.50 ↓	0.57 ↓	**0.66** ↑	0.41 ↓

Table 4. Multiclass classification (AD vs. MCI vs. HC) results comparing Sparse and Dense text representations and classifiers (DT, RF, SVM, LR) across dataset variations. Values represent classification accuracy. Best results are in boldface. The arrows denote a performance improvement ↑ or degradation ↓ relative to the Monolingual setting.

Confusion matrices

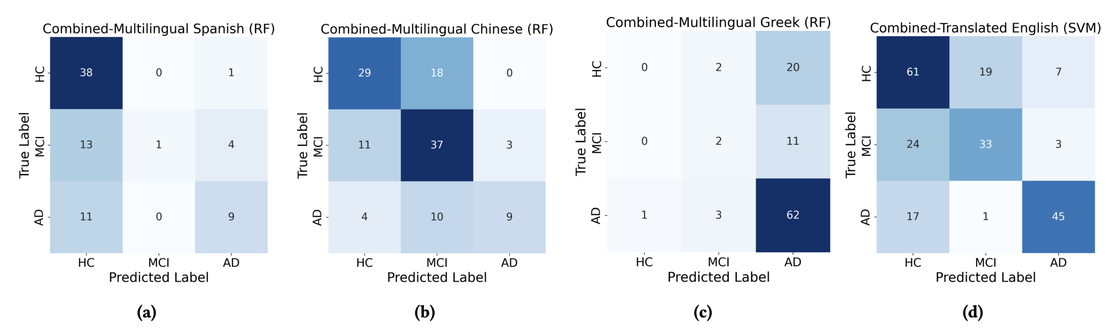

Figure 3: Confusion matrices for three-class classification using the best-performing dataset variant and classifier for each language.

Summary

Conclusion

- Binary classification outperformed multiclass due to MCI's clinical and cognitive overlap with AD and HC.
- Chinese improved with combined-multilingual (↑ 0.83→0.86) and combined-translation (↑ to 0.90) dataset variations; Spanish showed small gains, English mixed, and Greek declined.
- Translation helped Chinese and English, but hurt or had no effect on Spanish and Greek, likely due to altered language-specific cues.

Challenges

- Speaker diarization remains unreliable due to noisy recordings.
- Dataset imbalance impacts classification, especially for MCI cases.
- Translation does not always preserve cognitive-linguistic signals.

Future work

- Incorporate speaker role detection for more targeted modeling.
- Develop language-specific models using interpretable linguistic analysis.
- Explore ensemble and hierarchical deep learning architectures.
- Address label imbalance using advanced data augmentation methods.

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